TUT - 9

1) $f(s, t, k) \rightarrow$ Shortest path with at Most Edges.

- Recurrence

→ Path uses at most $k-1$ edges

Same as $f(s, t, k-1)$ or

→ Path uses exactly k edeges

Take min. of both cases

Base case

1) $f(s, s, 0) = 0$ for all $s \in V$

2) $f(s, t, 0) = \infty$ for all $s \neq t \in V$

NOTES

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Algorithm

Funct^n Shortest Path (G, s, t)

$n \leftarrow |V|$

For each vertex v in V:

$dp[v][0] \leftarrow \infty$

$dp[s][0] \leftarrow 0$

For $k = 1$ to $n-1$

For each vertex v in V:

$dp[v][k] \leftarrow dp[v][k-1]$

For each edge $(u \rightarrow v)$ with weight w in E:

If $dp[u][k-1] \neq \infty$;

$dp[v][k] \leftarrow \min(dp[v][k], dp[u][k-1] + w)$

Return $(dp[t][n-1])$

Zindagi ki Yatra me Aapka Sachha Sathi

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2)

→ Interval DP / Bottom UP

Base case

$m[i][i] = 0$ for all $i = 1$ to k

$m[i][j] =$ min over all p for i to $j-1$

$\{ m[i][p] + m[p+1][j]$

$+ n[i] \cdot n[p+1] \cdot n[j+1] \}$

$m[i][p] \rightarrow$ left

$m[p+1][j] \rightarrow$ right

multiply matrices

NOTES

Algorithm

Funct^n Matrix (n[1 .... k+1])

For i = 1 to k

min[i][j] ← 0

For L = 2 to k

For i = 1 to k - L + 1

j ← i + L - 1

m[i][j] ← ∞

For p = i to j - 1

cost ← m[i][p] + m[p+1][j] + n[i]. n[p+1]. n[j+1]

If cost < m[i][j]

m[i][j] ← cost

s[i][j] ← p

NOTES

Return m[1][k], S

Funct^n Print (s, i, j)

If i = j

Print "M" + i

Else

Print "("

Print Optimals (s, i, s[i][j])

—— " —— (s, s[i][j]+1, j)

Print ")"

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3)

→ cases

1) $f(i,j) = f(i+1, j-1)$ when $s[i] = s[j]$

2) $f(i,j) = 1 + \min[f(i, j-1), f(i+1, j)]$

when $s[i] \neq s[j]$

$$f(i,j) \begin{cases} 0 & i > j \\ f(i+1, j-1) & s[i] = s[j] \\ 1 + \min(f(i, j-1), f(i+1, j)) & s[i] \neq s[j] \end{cases}$$

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"Ab3bd"

L=0

f[1][1]=0, f(2)(2)=0,

f(3)(3)=0, f(4)(4)=0

f(5)(5)=0.

Final DP table

i|j	1	2	3	4	5
1	0	1	2	1	2
2	–	0	1	0	1
3	–	–	0	1	2
4	–	–	–	0	1
5	–	–	–	–	0